Acute Response to Ketogenic Diet Consumption on Lipid Biochemical Profiles of Wistar Rats

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ABSTRACT

Ketogenic diets have become popular for weight loss and in treatment of many diseases but there are still concerns regarding its impact on metabolic diseases and cardiovascular health. To obtain more comprehensive data on how ketogenic diet may affect the challenges of metabolic syndromes, this study aimed to evaluate the effects of short-term consumption of ketogenic diet on lipid biochemical profiles of healthy rat. Male Wistar rats weighing 150 ± 50 grams were randomly divided into 3 groups as follow: Control (fed standard rat chow; n = 10 rats), acute fed ketogenic diet (n = 5) and sub-acute fed ketogenic diet (n = 5). Ketogenic diet was made by adding 50% animal fat to the standard rat chow. Feeding lasted for 3 and 7 days for acute and sub-acute groups respectively. At the end of experiment, body weights were determined and blood samples obtained for lipid profile estimation using standard laboratory procedures. The results showed that ketogenic diet intake for 3 days cause a significant increase ($p<0.05$) in serum cholesterol and low density lipoprotein compared with the control diet. Conversely, there were significant decrease ($p<0.05$) in serum triglyceride (3 days ingestion), cholesterol (7 days ingestion) and high-density lipoprotein cholesterol (7 days ingestion) compared with the control. These findings indicate that ketogenic diet ingestion may predispose to metabolic syndromes and increase the risk of cardiovascular diseases via inhibitory impact on high density liproprotein cholesterol.

Key words: Nutritional therapy, Ketogenic diet, Fat diet, Cardiovascular disease, Dyslipidemia.

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1. INTRODUCTION

Cardio-metabolic syndrome is a clustering of several risk factors including central obesity, diabetes mellitus, hyper-coagulability, hypertension, insulin resistance and micro-albuminuria[1,2]. Collectively, these risk factors increase the risk of developing cardiovascular disease (CVD) and metabolic disorders such as dyslipidaemia[1]. The resulting abnormal lipid levels contribute significantly to global health challenges[3]. Nutritional and dietary therapy, weight loss, exercise, and scientifically proven nutritional supplementation have been reported to be appropriate in the management of dyslipidaemia[3, 4]. Globally, an increasing proportion of people are consuming Western Diet characterized by a high intake of fat and refined sugar[5]. Diets such as these increase the risk of developing cardio-metabolic syndrome[5-7] by inducing sustained increases in plasma glucose, triglycerides and very low-density lipoprotein[8]. Diet was therefore seen as a major weapon in the fight against CVD due to its impact on the myriad of CVD risk factor[9].

The ketogenic diet is a food regimen consisting of a high fat concentration, relatively low carbohydrate and protein content[10]. According to Kosinski and Jornayvaz[11], ketogenic diet triggers elevated ketone bodies production from fat breakdown to produce energy. Several studies showed ketogenic diet has therapeutic benefits and as such has been recommended as a supplementary therapy for acne, cancer, polycystic ovary syndrome, respiratory distress[12] and as anticonvulsant therapy in people with epilepsy[13,14], maintaining mood stability for people with bipolar disorder, helpful for people with type 2 diabetes, and

reducing cholesterol levels in obese patients[15]. Because ketogenic diets are high-fat and there still exist the assumption that high dietary fat intake causes poor lipid profiles and elevated cardiovascular risk, it was suggested that it is medically liable to follow-up the serum lipids profile of consumers of ketogenic diets[16]. To obtain more comprehensive data on how ketogenic diet may affect the lipid profile, this study investigated the effects of short-term consumption of ketogenic diet on lipid biochemical profiles in healthy adult Wistar rats

2. MATERIALS AND METHODS

2.1. Major food materials

Ketogenic diet is high in fat and thus the fat material used in this study is lard gotten from animal fat and was locally prepared. The animal fat was gotten raw within Ekpoma and prepared in the Department of physiology laboratory 2 through heating following standard procedures. The ketogenic diet was formulated by adding 50% animal fat (lard) to the standard rat chow as previously reported by Riahi and Riah[17] but with some modifications.

2.2. Care and management of animals

Fifteen adult male Wistar rats of comparable weights (150 ± 50g) were used for this study. They were obtained from the Animal House of the College of Medicine, Ambrose Alli University, Ekpoma. The animals were housed in plastic cage with a wire screen top in the experimental holding site of the Department of Physiology, College of Medicine, Ambrose Alli University, Ekpoma. The animals were fed *ad libitum* with standard rat chow (Bendel feeds and flour meal Ewu, Nigeria), clean tap water, and allowed to acclimatize for fourteen days.

2.3. Grouping of experimental animals

This study is experimental in nature. After acclimatization, rats were randomly assigned into three (3) groups of equal rats. This was represented as follows.

Group 1 (control group; n = 10): Rats in this group were fed standard rat chow and clean water alone throughout the experiment and blood sample collected at 3 and 7 days.

Group 2 (Acute fed Ketogenic diet; n = 5): Rats in this group were fed ketogenic diet and clean water for 3 days.

Group 3 (Sub-acute fed Ketogenic diet; n = 5): Rats in this group were fed ketogenic diet and clean water for 7 days.

2.4. Experimental procedure

After acclimatization, rats in each group received corresponding treatments. The rats were allowed to feed on standard rat chow (control) or the formulated ketogenic diet (test-groups; formed by adding 50g animal fat to 50g rat chow) *ad libitum*. These procedures were repeated the subsequent days for 3 days and 7 days for the acute and sub-acute groups respectively.

2.5. Sample collection

At the end of diet feeding for each groups and after an over-night fast, the animals were euthanized via cervical decapitation and blood sample collected into plain bottle from the jugular vein. The blood in plain bottle was centrifuged at 3000 rpm for 10 minutes The centrifuged serum was decanted and taken to the laboratory for the estimation of serum lipid profile.

2.6. Sample analysis

The serum concentration of total cholesterol, triglycerides, and high density lipoprotein concentrations were determined using commercial enzyme-linked immune-sorbent assay (ELISA) kits while low density lipoprotein and very low density lipoproteins were calculated based on standard formular. The parameters were estimated from serum samples as per the procedure described in the

kit.The concentration of Very Low Density Lipoprotein and LDL-Cholesterol was calculated from the Fiedewald's formula: LDL = TC – HDL-Chol – TG/5 **(**Where TC = Total Cholesterol**,** HDL-Chol = High Density Lipoprotein Cholesterol, TG = Triglyceride; TG/5 = Very Low Density Lipoprotein)[18].

2.7. Data analysis

The mean ± standard error of mean was determined and the One-Way ANOVA (Bonferroni post-hoc-test) statistical test was performed using SPSS version 20 software. The value P < 0.05 was considered statistically significant. Results were then presented in suitable charts.

3. RESULTS

3.1. Effect of ketogenic diet on serum lipid profile of healthy adult Wistar rats

Figure 1 showed the effect of ketogenic diet on serum total cholesterol (a), triglyceride (b), low density lipoprotein (c) and high density lipoprotein (d) cholesterol levels of healthy adult Wistar rats.

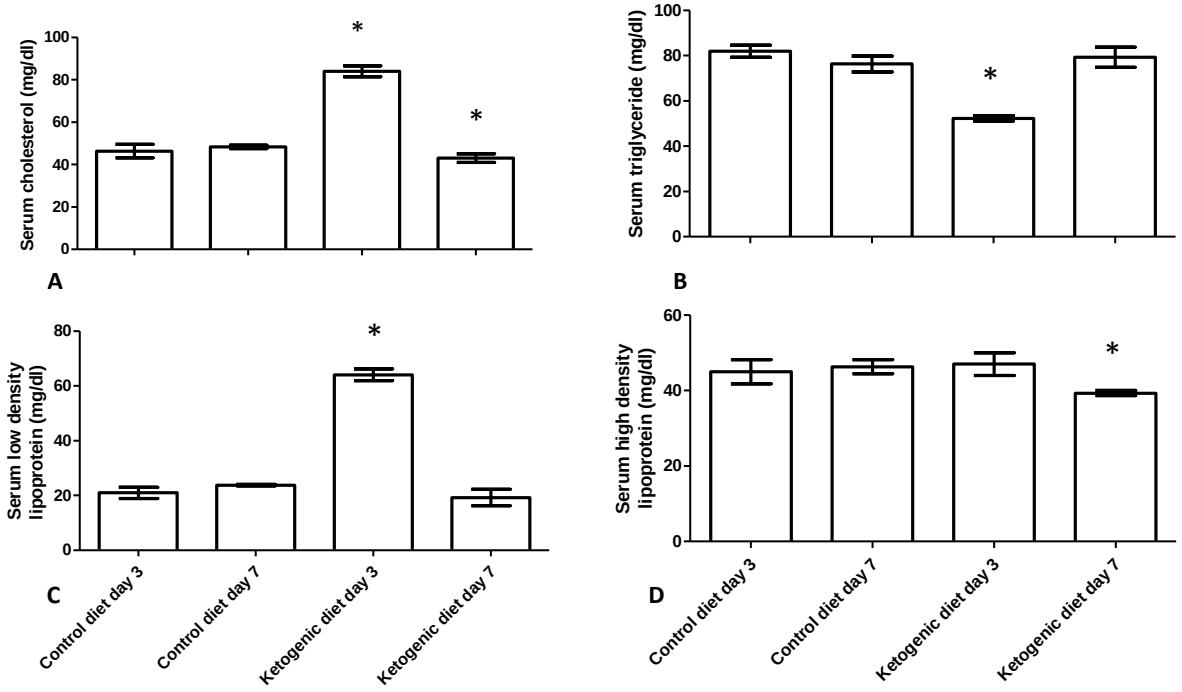


Figure 1. Effect of ketogenic diet on serum total cholesterol (a), triglyceride (b), low density lipoprotein cholesterol (c) and high density lipoprotein cholesterol (d) levels of healthy adult Wistar rats.

N= 20, n = 5. Values are in Mean ± standard error of mean; *statistically significant compared with control.

There was no significant difference in serum lipid variables between the 3 and 7 days of the control. However, ketogenic diet ingestion for 3 days resulted in significant higher serum total cholesterol level ($p<0.05$) compared to the controls while 7 days ingestion resulted in a significant decrease. Ketogenic diet ingestion for 3 days resulted in a significantly lower serum triglyceride level ($p>0.05$) compared to the control and ketogenic diet ingestion for 7 days. Ketogenic diet ingestion for 3 days resulted in a significantly higher serum low density lipoprotein cholesterol level ($p>0.05$) compared to the control and

ketogenic diet ingestion for 7 days. Ketogenic diet ingestion for 7 days resulted in a significantly lower serum high density lipoprotein cholesterol level ($p>0.05$) compared to the controls and ketogenic diet ingestion for 3 days.

4. DISCUSSION

Ketogenic diets have been criticized on the grounds that they jeopardize health[19]; however, very few studies have directly evaluated its effect on risk factors for CVD[9]. While caloric restriction or other form of reduction in nutrient can reduce risk of CVD and dyslipidemia, manipulating dietary macronutrient content has been shown to improve blood lipid profiles even in the absence of weight loss[20]. In the present study, ketogenic diet induced various variable changes in serum cholesterol, triglyceride, LDL and HDL at acute and sub-chronic ingestion. Specifically, the 3 days ingestion of ketogenic diet caused an increase in serum cholesterol and LDL while serum triglyceride decreases with an unchanged HDL. However, following 7 days of ketogenic diet ingestion there was reduction in serum cholesterol, LDL and HDL with an unchanged triglyceride level. These findings indicate that acute intake of ketogenic diet may be more deleterious to the cardiovascular system as compared to sub-chronic intake. However, the decrease in HDL by ketogenic diet is not favourable. In comparison to the present study, high-fat load for 30 days have been documented to result in increased levels of total cholesterol, triglyceride, LDL and cholesterol, in the rats' blood serum[21]. Also, saturated fats have been reported to modify cholesterol metabolism and increase significantly the concentration of HDL cholesterol and a raised triglyceride accumulation in mouse after three weeks of high fat diet[22]. The difference in the findings of the present study with that of Nangue et al.[22] may be due to the difference in duration of exposure to the diet between both studies. However, other studies indicated that application of ketogenic diet results in significant decrease in serum triglycerides, small increase in total and LDL cholesterol and moderate increase in HDL cholesterol in normal lipid profiled human subjects[23].

The data from our study demonstrate that ketogenic diet exerts a distinctly pronounced hyperlipidemic effect on serum cholesterol and LDL following acute ingestion as compared to sub-acute ingestion. S everal animal models have been used to learn about the effect of a high-fat diet on the function of vital organs, such as the kidneys and liver[24,25]. High-fat diet is more likely to trigger a reduction in mitochondrial quinone pool and is associated with increased mitochondrial reactive oxygen species (ROS) formation in the rat liver[26]. Thus, high fat diets which in the case of this study; a ketogenic diet, may alter liver metabolism and in turn induce changes in lipid profile. In support of this assertion, a high-fat diet has been shown to induce alteration in renal lipid metabolism in mice, especially the balance between lipogenesis and lipolysis, leading to the accumulation of lipid in the kidneys and, consequently, renal dysfunction[27].

5. CONCLUSION

This study showed that short term intake of ketogenic diet alters lipid profile in adult male Wistar rats. Specifically, sub-chronic ingestion may lead to cardiovascular rick via serum HDL inhibition with acute intake stimulating serum LDL. These findings suggest that ketogenic diet may increase the risk of cardio-metabolic syndrome. It is therefore recommended that intake of ketogenic diet be monitored by a health care specialist and there is a need for further studies.

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